



Research article

Failure to thrive: A QuantCrit analysis of academic failure and everyday discrimination in undergraduate nursing education

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ABSTRACT

Background: Academic failure in nursing education is frequently framed as a student deficit, detached from the structural and institutional forces that shape educational outcomes. This framing obscures how racism, bias, and exclusionary evaluation practices influence who fails—and under what conditions.

Objective: This study examines the relationship between everyday discrimination and academic failure in undergraduate nursing education, using a Quantitative Critical Race Theory (QuantCrit) lens to foreground identity as a site of structural vulnerability.

Methods: A cross-sectional survey was conducted with 256 undergraduate nursing students at a Canadian university. Descriptive analyses, chi-square tests, *t*-tests, and logistic regression models were used to examine associations between everyday discrimination, academic failure, and social identity.

Results: Students who reported academic failure had higher levels of perceived everyday discrimination. Academic failure was more common among racialized students (54.4 %) compared to non-racialized students (38.5 %). The interaction of race and gender revealed that racialized women had over seven times the odds of academic failure compared to non-racialized men. Skills-based assessments were the most common site of reported failure.

Conclusions: These findings challenge the notion that student failure is solely due to deficits. Instead, they suggest that structural inequalities, particularly those based on race and gender, significantly impact academic outcomes. A QuantCrit perspective redefines failure as a consequence of institutional structures and power dynamics, influencing evaluation practices, faculty development, and equity accountability in nursing education.

1. Introduction

Despite increased attention to equity, diversity, and inclusion in nursing education, academic failure continues to fall unevenly across student populations. Racialized, Indigenous, and other equity-deserving students remain disproportionately affected, yet these patterns are rarely examined as outcomes of structural forces. In this study, *racialized* refers to people who are socially constructed and positioned as non-white within dominant power structures—a framing that emphasizes race as a social process rather than a biological category, consistent with Canadian human rights usage (Ontario Human Rights Commission, 2017). Nevertheless, failure is still more often explained as an individual deficiency, attributed to study skills, resilience, or motivation, while the institutional conditions that disadvantage students remain unexamined and intact (Oducado, 2021; Pacheco-Castillo et al., 2021; Russell et al.,

2022).

This framing is particularly troubling in the Canadian context. The Truth and Reconciliation Commission (TRC) of Canada's *Calls to Action* (Truth and Reconciliation Commission of Canada, 2015) urged institutions in both the education and health sectors to address colonial legacies and promote meaningful reconciliation. Nursing, positioned at the intersection of both, has a unique responsibility to lead these efforts. Yet institutional responses have largely focused on increasing Indigenous student enrollment or embedding curriculum content on colonization and cultural safety, often without addressing the deeper structural inequities that shape who succeeds and who fails (Lane and Petrovic, 2018; Pijl-Zieber and Hagen, 2011).

While the proportion of Indigenous students entering nursing programs has increased, graduation rates remain low (Exener-Pirot, 2018). Institutional data practices contribute to this gap in understanding:

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Indigenous students are often tracked only if they enter through access programs or self-identify under special consideration, while many choose not to disclose their ancestry to avoid surveillance and discrimination (Settles et al., 2019; Slatyer et al., 2016; Van Bower et al., 2021). This creates a fragmented and pathologizing picture, one that reinforces stereotypes of Indigenous vulnerability (Henschke, 2017), intellectual inferiority (Caskill, 2012), and deviancy (de Leeuw et al., 2010), while failing to capture the broader impacts of racism in nursing education. These same deficit narratives have long justified surveillance and intervention in health and child welfare systems (Beavis et al., 2015; de Leeuw et al., 2010), and similar logics may shape educational evaluation.

Though data on the surveillance of Indigenous nursing students is limited, existing research in K–12 education has documented the disproportionate discipline and monitoring of Indigenous students (Ontario Human Rights Commission, 2017). Such dynamics obscure the historical and political forces that produce educational disparities and conceal institutional complicity in their reproduction (Donald, 2018; Manhire-Heath et al., 2019). Moreover, in the absence of disaggregated data, disparities for other marginalized groups, such as Black, Latino, and gender non-conforming students, remain unexamined, despite well-documented inequities in other international educational contexts (Ackerman-Barger et al., 2020; Gillborn et al., 2018; Henry et al., 2021; Owens and McLanahan, 2020). Intersectional patterns, where multiple social identities combine to exacerbate inequity, are also obscured by this lack of comprehensive data (Brockman, 2021).

In response, this study adopts a Quantitative Critical Race Theory (QuantCrit) framework to interrogate how everyday experiences of discrimination relate to academic failure in undergraduate nursing education. QuantCrit challenges assumptions of objectivity in conventional quantitative research and repositions identity not as a confounder, but as a site of structural meaning (Castillo and Babb, 2024; Gillborn et al., 2018). We approach data as socially constructed and embedded in systems of power, recognizing that bias can shape all aspects of the research process, from question formation to interpretation (Hawn Nelson and Zanti, 2020). Our aim is not only to identify patterns of inequality but to resist reproducing them through methodological choices. This inquiry centres the voices and experiences of those who bear the greatest burden of academic failure and positions research as a tool for equity and accountability.

2. Background

Academic failure in undergraduate nursing education is frequently explained through deficit-based models that emphasize students' lack of preparedness, resilience, or motivation (Bulfone et al., 2021; Oducado, 2021; Pacheco-Castillo et al., 2021). These accounts dominate both research and institutional practice, locating the problem within the individual and deflecting attention from the educational systems that shape student outcomes. Even when psychological distress or remediation challenges are acknowledged, the focus tends to remain on internal shortcomings rather than on the broader structures in which students are embedded (Leon, 2020; Lewis et al., 2022).

A substantial body of research has documented how racism, institutional bias, and structural exclusion shape the learning environments of racialized, Indigenous, and other equity-deserving students. Reports of microaggressions, biased assessments, curricular erasure, and tokenization are common, compounded by the absence of racialized faculty, culturally responsive pedagogy, and safe spaces for belonging (Beagan et al., 2023; Iheduru-Anderson et al., 2020; Luhanga et al., 2023). These experiences are not incidental; they are embedded in the normative structures of nursing education, which continue to reflect and reproduce white, middle-class values and standards (Ackerman-Barger et al., 2020; Bell, 2021).

These dynamics are particularly evident in the evaluative landscape of nursing education. Clinical and skills-based assessments, especially

those requiring performance or demonstration, are often steeped in implicit expectations around communication, affect, and professionalism. These standards may be unevenly accessible across social locations, particularly when they are unstated, subjective, or grounded in dominant norms (FitzGerald and Hurst, 2017; González et al., 2018). Racialized and Indigenous students may face both hyper-surveillance and invisibility in these settings, producing conditions in which failure is not simply personal but structural (Van Bower et al., 2021; Zanchetta et al., 2021).

Faculty play a central role in shaping these environments but often feel unprepared to engage with racial inequity in meaningful ways. Discomfort with the topic, limited training, and a lack of institutional support contribute to a climate of avoidance, further marginalizing students and curtailing opportunities for structural change (Bennett et al., 2019; Gordon and Patterson, 2023). Institutional equity initiatives, though well-intentioned, often reinforce the logics they aim to disrupt. Rather than interrogating the systems that produce unequal outcomes, such efforts tend to focus on bolstering individual coping or remediation (Pijl-Zieber and Hagen, 2011; Sedgwick et al., 2025). This positioning renders students visible primarily through lenses of deficit and risk, while leaving structural power unchallenged.

Although calls for anti-racist and decolonial approaches to curriculum and faculty development have grown louder (Ackerman-Barger et al., 2020; Bell, 2021; Ramsay-Seaner et al., 2022), few studies have examined how these structural conditions map directly onto measurable academic outcomes like course failure, particularly using quantitative methods. Most statistical studies treat identity as a confounder, not a consequence of power. As a result, the patterned nature of academic failure across racial, gendered, and intersectional lines remains obscured. To address this gap, our study reframes academic failure not as a neutral or merit-based outcome, but as a structural indicator of institutional inequity. This requires a shift not only in what we study, but in how we study it.

This study is guided by Quantitative Critical Race Theory (QuantCrit), a methodological orientation that extends the insights of Critical Race Theory (CRT) into the realm of quantitative research. QuantCrit challenges the assumptions of objectivity, neutrality, and universality that often underlie traditional statistical approaches. These assumptions can obscure structural inequality and reproduce dominant power relations (Castillo and Babb, 2024; Gillborn et al., 2018; Vargas and Peet, 2024). Conventional research often treats race, gender, and other identities as confounding variables, reinforcing narratives of individual failure while masking structural patterns (Gillborn et al., 2018). In contrast, QuantCrit treats these identities as structurally meaningful: not just descriptive categories, but as markers of power, exclusion, and social location. Our use of QuantCrit is grounded in the recognition that the questions researchers ask, the ways they collect and interpret data, and the choices they make in analysis are shaped by their own positionalities and the institutions in which they work (Hawn Nelson and Zanti, 2020).

Gillborn et al. (2018) outline five core tenets of QuantCrit, which inform our approach:

1. The Centrality of Racism: Racism is not an aberration but a central organizing feature of social institutions, including education.
2. Numbers Are Not Neutral: Quantitative data are shaped by the contexts in which they are collected, categorized, and interpreted.
3. Categories Are Socially Constructed: Identity categories (such as "race") are neither natural nor fixed; they are produced through historical and political processes.
4. Data Require Context and Voice: Data do not speak for themselves. They must be interpreted through critical engagement with the experiences and perspectives of marginalized groups.
5. Quantitative Research Must Serve Social Justice: Data should be used to expose inequities and support systemic change, not simply describe patterns.

In applying these tenets, we resist framing identity as something to be statistically neutralized. Instead, we ask how identity functions within systems of nursing education: how it structures exposure to discrimination; how it shapes evaluative interactions; and how it may be implicated in institutional mechanisms of failure. Our goal is not to isolate causal effects, but to trace patterns of inequities and raise critical questions about the systems that produce them.

This study emerged in part from the observation that disaggregated outcome data are rarely available for nursing education in Canada. While Canadian universities increasingly invite Indigenous students to self-identify, these data are not routinely shared with faculties. Nor are there consistent practices for collecting data on racialized, gender-diverse, disabled, or financially insecure students (Settles et al., 2019; Slatyer et al., 2016). The result is a narrow and often pathologizing picture, one that over-represents Indigenous students through special admission programs while rendering others invisible. QuantCrit offers a framework for challenging these limitations and interpreting data in ways that foreground power, history, and structure.

Our analysis draws on this framework to examine how experiences of everyday discrimination relate to academic failure in nursing education. We centre identity and discrimination not as background variables, but as analytical priorities that shape both our methods and our interpretations of what counts as evidence, and for whom.

2.1. Researcher positionalities

This research was carried out by a collective of scholars with varied social locations and relationships in nursing and health professions education. Our team includes individuals who identify as Indigenous, racialized, settler, and linguistically diverse. We are also educators and former students who have, in different ways, experienced the effects of academic failure and discrimination within post-secondary institutions. These experiences do not stand outside the research. They shape how we understand the problem, how we interpret the data, and how we define what counts as failure in the first place.

We come to this work with a shared commitment to challenging the systems that reproduce inequity in nursing education. Our interpretations are not objective in the conventional sense, but situated, relational, and shaped by ongoing critical reflection. We recognize that positionality is not a disclaimer. Positionality is part of our method. Our approach is grounded in the belief that knowledge production is never neutral, and research can either reinforce or interrupt structural harm. We choose the latter.

3. Methods

3.1. Design

This study draws on survey data from a larger mixed-methods project grounded in the principles of Critical Participatory Action Research (CPAR) (Fine et al., 2021). The broader project involved both quantitative and qualitative components, including cross-sectional survey responses and sharing circle conversations with nursing students. While the integration of these data sources is discussed elsewhere, this manuscript focuses specifically on a subset of the quantitative findings. Here, we examine responses to the Everyday Discrimination Scale (EDS) (Williams et al., 1997) in relation to students' reported experiences of academic failure. Our aim is not only to document patterns, but to question the institutional conditions that may produce and sustain them.

3.2. Participants and setting

Undergraduate nursing students from a single institution in Canada were recruited via institutional email lists. To be eligible, participants had to be actively enrolled in the Bachelor of Nursing (BN) program and provide informed consent. In Summer 2023, a total of 690

undergraduate nursing students were enrolled. Of these, 280 students completed the survey, resulting in a response rate of approximately 40.6 %. Aligned with our QuantCrit orientation, all students were invited to participate regardless of identity or academic standing. As institutional demographic data on race, other than Indigenous self-identification, are unavailable, we were unable to assess sample representativeness. Rather, our study illuminated the demographics of the undergraduate nursing student population at our institution in previously unexplored and unaccounted ways.

3.3. Data analysis

This analysis was guided by Quantitative Critical Race Theory (QuantCrit), which challenges assumptions of objectivity and neutrality in quantitative research and repositions data analysis as a tool for interrogating structural inequality (Castillo and Babb, 2024; Gillborn et al., 2018). Informed by the systemic absence and selective pathologization of equity-deserving students in institutional data practices, we intentionally foregrounded identity variables not as confounders, but as structurally meaningful indicators of power, exclusion, and social location.

Our analytic approach prioritized disaggregation, consistent with both QuantCrit principles and calls for data practices that reveal, rather than obscure, inequities (United Nations, 2018). We began by generating descriptive statistics to examine the distribution of Everyday Discrimination Scale (EDS) scores and the proportion of students who reported academic failure. These distributions were disaggregated by key demographic variables, including race, gender, age, and financial status, in order to surface differences across identity groups.

To assess relationships between identity and academic failure, we conducted a series of chi-square tests for categorical variables and used independent-sample *t*-tests and one-way ANOVAs to explore differences in mean EDS scores across identity groups and between students who had and had not experienced failure. These bivariate analyses helped identify patterns of marginalization and exclusion.

To examine associations more directly, we constructed logistic regression models using EDS scores as the primary independent variable and academic failure (yes/no) as the outcome. Additional models included race, racialization, gender, and interaction terms (e.g., Race $\times$ Gender) to explore compounding effects. In keeping with our QuantCrit orientation, identity categories were not treated as statistical confounders to be neutralized, but as markers of students' structural location within educational systems.

We initially disaggregated racial identity to examine differences between specific groups (e.g., Black, Indigenous, Asian, white). However, due to small cell sizes and a lack of statistical significance in some disaggregated comparisons, through discussion and as a team, we made the analytic decision to collapse these identities into a broader "racialized" category, compared to white students. This decision was guided by the underlying patterns observed in the disaggregated data, which, while not meeting conventional significance thresholds, consistently indicated disparity. We recognize that collapsing flattens important differences across groups. However, we approached it not as a retreat from disaggregation, but as a strategic decision to retain visibility of structural inequality given the limitations of statistical power. Our use of the racialized category was not to mask difference, but to reflect it, and to resist the erasure of inequity through overreliance on arbitrary statistical thresholds (Gillborn et al., 2018).

For transparency, descriptive statistics for disaggregated racial identity groups are provided in Supplementary Table 1. While statistical comparisons between specific groups were underpowered, the table allows readers to observe the direction and consistency of disparities prior to our analytic decision to collapse into a broader 'racialized' category.

All analyses were conducted in SPSS (v28). In addition to *p*-values, we report effect sizes and 95 % confidence intervals to support transparent and meaningful interpretation. Our goal was not simply to

identify statistically significant outcomes, but to trace patterns that reflect institutional conditions and raise questions about structural accountability.

Missing data were present, particularly within the Everyday Discrimination Scale. Some students skipped one or more EDS items, while 18 submitted blank or near-blank surveys but completed the submission process and received the honorarium. These incomplete responses were excluded from analysis. However, rather than treating missingness as mere error, we interpret it as part of the research context. Non-response may reflect discomfort, disengagement, or distrust, particularly in a study that asks students to name their experiences within systems that may have previously harmed them. In this framework, silence is not absence; it is a form of data that reflects the broader institutional landscape of power and safety in which students are situated.

3.4. Ethics

Ethics approval for this study was initially granted by the university’s Research Ethics Board on May 1st, 2023 (HE 2023-0059). Informed consent was obtained electronically before participation began. Given the possibility that students from small or multiply marginalized groups could be indirectly identifiable, particular care was taken to protect confidentiality. The survey instrument was co-developed with student researchers from equity-deserving communities, who also contributed to the interpretation of data and validation of findings. This collaborative approach reflects our commitment to ethical research practices that prioritize inclusion, accountability, and respect—not only in the analysis, but throughout the research process itself. Participants received a \$40 gift card honorarium in recognition of their time. This amount was chosen to acknowledge participants’ contributions while minimizing the risk of undue influence, consistent with institutional ethics guidelines.

4. Results

4.1. Sample characteristics

A total of 280 students participated. The sample was diverse, with 70.1 % identifying as racialized (Black, Indigenous and Person of Colour) and 21.6 % as white. Disability was reported by 9.6 %, financial insecurity by 42.5 %, and 41.2 % were EAL speakers. Most participants identified as women (87.3 %), heterosexual (86.8 %), and were under 27 years of age (84.3 %). Students were distributed across all years of the program (Table 1).

Table 1
Participant social identities.

Social identity	n (%)
Racialized	185 (70.1)
Asian	150 (56.8)
Indigenous	23 (8.7)
Black	34 (12.9)
White	57 (21.6)
Reported disability	24 (9.6)
Financial insecurity	102 (42.5)
English as an Additional Language (EAL)	106 (41.2)
Woman	214 (87.3)
Man	31 (12.7)
Heterosexual	203 (86.8)
2SLGBTQ+	31 (13.2)
Age under 27 years	214 (84.3)
Age 27 years and older	40 (15.7)
Year 2 enrollment	86 (34.3)
Year 3 enrollment	91 (36.3)
Year 4 enrollment	74 (29.5)

Note. Ns vary due to missing data; percentages may not total 100 % due to rounding.

Of the full sample, 256 participants provided sufficient data to be included in the analysis. To be included, students needed to complete the academic failure item and respond to at least five items on EDS, allowing for the calculation of a mean EDS score in line with established scoring protocols (Williams et al., 1997). The remaining 24 participants were excluded: some did not meet the minimum response threshold for the EDS, others skipped the failure item, and a small number submitted the survey without responding to any questions. Although these participants received the honorarium, their responses were not included in the analytic sample.

Of the 262 participants included in the analysis, 50 % (n = 131) reported having experienced academic failure in at least one course—whether in theory, clinical, or skills-based—during their nursing education.

4.2. Course type and failure

Among students who reported academic failure, the most common course type was skills-based. Fifty percent of students (n = 131) reported failing at least one skills demonstration during their nursing program. Most of these students passed the evaluation on a subsequent attempt. Only 2.3 % of the total sample (n = 6) reported failing a skills evaluation on their final attempt. Theory course failures were reported by 6.8 % of students (n = 18), while failure in clinical courses were rare (n = 2; <1 %).

4.3. Everyday discrimination and academic failure

Students who reported academic failure had significantly lower mean EDS scores (M = 3.94) than those who did not (M = 4.15), $t(256) = 2.57, p < .05$, Cohen’s $d = 0.17$. Since lower EDS scores reflect more frequent discrimination, this indicates that students who experienced academic failure also reported more frequent everyday discrimination.

4.4. Attributions of discrimination

Among participants who completed the EDS, the most selected attribution for perceived discrimination were race (26.0 %), followed by gender (17.0 %), age (16.4 %), and physical appearance (13.6 %).

4.5. Academic failure by social identity

Failure rates were broadly similar across most social identity groups. Women and men reported failure at comparable rates (50.2 % vs. 46.7 %; OR = 0.867, 95 % CI: 0.403–1.864). Two-Spirit, Lesbian, Gay, Bisexual, Transgender, Queer or Questioning, and others (2SLGBTQ+) students reported slightly higher rates of academic failure (56.7 %) than cis-gendered, heterosexual peers (51.5 %; OR = 0.812, 95 % CI: 0.375–1.175).

Odds of academic failure also varied by racial identity. Compared to white students, Black students had 58.8 %, OR = 1.96, 95 % CI: 0.83–4.65, Asian students had 51 %, OR = 1.43, 95 % CI: 0.77–2.66, and Indigenous students had 45.5 %, OR = 1.15 (95 % CI: 0.43–3.09). While these group comparisons were not statistically significant, descriptive trends suggest that Black, Asian and Indigenous students had higher rates of failure than white students. And, racial identity once collapsed into a binary (racialized vs. non-racialized) was significantly associated with failure. Among racialized students, 54.4 % reported academic failure compared to 38.5 % of non-racialized students, $\chi^2(1, N = 260) = 5.55, p = .019$ (Table 2).

Although not statistically significant, higher failure rates were observed among students with disabilities (54.2 % vs. 50.7 %; OR = 0.87, 95 % CI: 0.37–2.02), those experiencing financial insecurity (53 % vs. 51 %; OR = 0.93, 95 % CI: 0.56–1.56), and students who reported EAL (54 % vs. 48 %; OR = 0.79, 95 % CI: 0.48–1.31) compared to their peers.

Table 2
Experience of academic failure by racialized identity (N = 260).

Experience of failure	Non-racialized, n (%)	Racialized, n (%)	Total
No	47 (61.5)	83 (45.6)	131
Yes	30 (38.5)	99 (54.4)	129
Total	78 (100)	182 (100)	260

Note. Percentages are within each racialized group. $\chi^2(1, N = 260) = 5.55, p = .019$.

4.6. Interaction of race and gender

The social identities reported in the participant characteristics such as race, gender, language, disability status, and financial insecurity, served as the basis for our regression models and interaction terms. While we explored multiple combinations of these identity markers, only one interaction term reached statistical significance: the interaction between race and gender. Specifically, racialized women were over seven times more likely to report academic failure compared to non-racialized men (OR = 7.21, 95 % CI: [1.01–51.34], $p = .05$). While the wide confidence interval reflects small subgroup sizes, the magnitude and direction of this effect are instructive, consistent with QuantCrit, which highlights how even underpowered analyses can reveal compounded structural disadvantage.

5. Discussion

This study contributes to a growing body of scholarship that challenges individual-deficit models of academic failure in nursing education. While mainstream literature frequently attributes failure to poor preparation, lack of resilience or self-efficacy (Oducado, 2021; Pacheco-Castillo et al., 2021; Xu et al., 2023), our findings point to structural explanations that more adequately account for who fails, under what conditions, and to what effect. Students who reported experiencing academic failure were significantly more likely to report everyday discrimination, particularly racialized students. These patterns suggest that exclusion in nursing education is not merely peripheral, but structurally embedded.

Prior scholarship has illustrated how racialized and equity-deserving students routinely navigate educational environments shaped by institutional bias, cultural misrecognition, and limited access to support (Beagan et al., 2023; Iheduru-Anderson et al., 2021). Our findings affirm and extend this work by linking those experiences to a specific and measurable outcome: failure in evaluative contexts. Notably, students most often reported failure in skills-based assessments. Unlike theory-based evaluations, which may offer clearer rubrics and more transparent grading schemes, clinical and skills demonstrations are highly performative, context-dependent, and susceptible to subjectivity (González et al., 2018). Expectations around professionalism, competence, and communication often remain implicit (Edwards-Maddox et al., 2022; FitzGerald and Hurst, 2017), and students are expected to perform them fluently and without error. Such assessments are not neutral. They carry the weight of institutionalized subjectivities that disproportionately disadvantage racialized and Indigenous students, reproducing long-standing inequities under the guise of meritocratic evaluation (Sokolowich, 2025).

This evaluative terrain can be particularly fraught for racialized and Indigenous students, who encounter both hyper-surveillance and invisibility within institutional spaces (Ray King et al., 2021; Van Bower, 2022; Zanchetta et al., 2021). As our findings suggest, these dynamics may culminate in academic failure not as an isolated event, but as the outcome of sustained exposure to marginalization and misrecognition. That students most frequently attributed discrimination to race, and that such attributions were significantly associated with failure, reinforces the argument that racism in nursing is not merely interpersonal but institutional in its effects.

Notably, our regression analysis revealed a significant interaction between race and gender: racialized women had over seven times the odds of experiencing academic failure compared to non-racialized men. While other interaction terms across social identity categories did not reach statistical significance, this result underscores how marginalization is compounded at the intersection of racialization and gender. These findings affirm Crenshaw’s (1991) theory of intersectionality, which emphasizes that systems of power operate through multiple, overlapping forms of exclusion. In this context, racialized women face a form of structural violence that is produced and sustained by institutions that fail to see or account for the particularities of their experiences. The significance of this interaction lies not merely in its statistical outcome but in what it reveals about how institutions disproportionately disadvantage those situated at multiple axes of exclusion, even in professions such as nursing, where women are numerically dominant.

Crucially, our QuantCrit lens enabled us to retain interaction terms and treat non-significant results not as evidence of equity but as signal of structural dynamics that may be underpowered, obscured, or methodologically constrained (Castillo and Babb, 2024; Gillborn et al., 2018). While disaggregated comparisons between individual racial identity groups and white students did not consistently reach statistical significance, the trends were suggestive of disparity. These patterns informed our decision to collapse racialized identities into a single analytic category, ensuring inequities were not rendered invisible by limited power. Similarly, other identities such as financial status, EAL, and disability showed non-significant but consistent trends that shaped our interaction models—some of which revealed statistically significant patterns of compounded discrimination. In this way, our analysis underscores that identity is not a random variation to be neutralized, but a structurally meaningful site of exclusion and accountability.

Survey patterns also prompted critical reflection. A subset of students declined to respond to items on discrimination or academic failure or submitted incomplete surveys. While excluded from quantitative analysis, these silences carry meaning. As others have noted, non-response in equity-focused research may reflect fatigue, fear, or distrust, particularly among students who have experienced institutional harm (Gillborn et al., 2018). Within a QuantCrit orientation, silence is not absence, but a data point that warrants ethical and interpretive consideration.

Taken together, these results highlight how discrimination and structural bias are embedded in the evaluative practices of nursing education. Skills-based assessments emerged as a key site of failure, suggesting that definitions of competence and professionalism are themselves raced, gendered, and classed. The consistent trends across racial, socioeconomic, linguistic, and disability categories point to overlapping axes of exclusion, even where statistical power was limited. These findings set the stage for our implications, underscoring that reforming evaluative practices is not only a pedagogical concern but also a matter of equity accountability for nursing education.

5.1. Implications

The findings of this study carry important implications for nursing education, particularly for programs committed to equity, diversity, and inclusion. First, the persistent framing of academic failure as an individual deficit must be critically re-evaluated. Our results demonstrate that failure is significantly associated with experiences of discrimination, and that racialized and equity-deserving students, particularly racialized women, are more likely to report both exclusion and academic difficulty. These patterns compel a shift in focus: from what students lack to how institutions fail.

The association between skills-based assessments and reported failure underscores the need to scrutinize how evaluation itself functions as a site of institutional bias. These assessments are not neutral; they often rely on implicit norms around communication, demeanor, and “professionalism” that are deeply raced, classed, and gendered. Nursing programs must interrogate these evaluative assumptions. Revising rubrics

or offering faculty training, while useful, are insufficient unless accompanied by a critical reckoning with how standards are constructed, by whom, and for whom. Programs should develop transparent, collaboratively created criteria for skills-based evaluations and ensure students can question or appeal judgments without fear of reprisal. Faculty development must also go beyond generic equity training to engage explicitly with how evaluative bias is enacted and how it can be interrupted.

At an institutional level, routine practices of controlling for identity in research and program evaluation must be replaced by disaggregated, equity-oriented approaches that illuminate structural patterns. Yet data alone is not enough. Institutions must act on the disparities that become visible—by diversifying faculty, embedding anti-oppressive pedagogies, and developing governance mechanisms that include and are accountable to equity-deserving students.

Finally, nursing education must fundamentally reframe failure. It should not be seen as a neutral outcome of academic merit or individual shortcoming. Rather, failure must be recognized as a symptom of misalignment between institutional structures and the diverse students they serve. Addressing this misalignment requires not only inclusion, but transformation—of what counts as success, how it is evaluated, and who is supported to thrive.

6. Limitations

This study was conducted at a single site, and demographic data for the full student population were unavailable, limiting assessment of representativeness. Based on disaggregated patterns, some identity categories were collapsed to ensure analytic power. We do not treat missing responses as error. We acknowledge that these absences may be a meaningful expression of student distrust or discomfort vis-à-vis our research and nursing education more broadly. Future research should extend this work through multi-site studies with larger and more diverse samples, as well as longitudinal designs to examine how structural conditions and external accountability pressures shape student trajectories.

7. Conclusion

Academic failure in nursing education is too often dismissed as a personal shortcoming; framed as something to be addressed through student support, resilience-building, or remediation. Our findings challenge this framing by revealing how experiences of everyday discrimination are associated with academic failure, particularly among racialized and equity-deserving students. These outcomes are not random. They are structured.

Using a QuantCrit lens, we treated identity not as a statistical confounder, but as a meaningful indicator of how institutions distribute opportunity, risk, and harm. This approach made visible the compounded vulnerability of racialized women in evaluative settings and revealed how failure may function as a mechanism of institutional exclusion. To meet the stated goals of equity in nursing education, institutions must move beyond superficial interventions. They must critically examine how core practices, such as assessment, professionalism, and evaluation, are shaped by whiteness, gender norms, and institutional power.

Our study offers one model for how equity-oriented, quantitative research can support that work: not by neutralizing identity, but by naming it as central to how success and failure are produced. While our study is not exhaustive, it is instructive. It highlights the need for deeper accountability and systemic change in how nursing education defines, evaluates, and supports student success, not in spite of difference, but through a commitment to justice.

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.nedt.2025.106885>.

CRediT authorship contribution statement

Vanessa Van Bower: Writing – review & editing, Writing – original draft, Supervision, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Conceptualization. **Abraham Mekonnen:** Visualization, Validation, Software, Investigation, Formal analysis. **Marnie Kramer:** Writing – review & editing, Funding acquisition, Conceptualization.

Consent to participate

All participants provided informed electronic consent prior to survey and sharing circle participation.

Consent for publication

Not applicable.

Ethical considerations

Ethical approval was granted by University of Manitoba's Research Ethics Board 1 (HE2023–0059). Indigenous ethical principles of respect, reciprocity, relevance, and relationships were integral to the research design and execution.

Declaration of Generative AI and AI-assisted technologies in the writing process

During the preparation of this manuscript, the authors used ChatGPT (OpenAI, GPT-4) to assist with grammar, clarity, and organization of language. The AI was not used for data analysis, content generation, or interpretation. All outputs were reviewed and edited by the authors, who take full responsibility for the content of this manuscript.

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Research data supporting this publication are available upon reasonable request to the corresponding author, subject to ethical and privacy considerations.

References

- Ackerman-Barger, K., Boatright, D., Gonzalez-Coloso, R., Orozco, R., Latimore, D., 2020. Seeking inclusion excellence: understanding racial microaggressions as experienced by underrepresented medical and nursing students. *Acad. Med.* 95 (5), 758–763. <https://doi.org/10.1097/ACM.0000000000003077>.

- Beagan, B.L., Bizzeth, S.R., Etowa, J., 2023. Interpersonal, institutional, and structural racism in Canadian nursing: a culture of silence. *Can. J. Nurs. Res.* 55 (2), 195–205. <https://doi.org/10.1177/08445621221110140>.
- Beavis, A.S.W., Hojjati, A., Kassam, A., Choudhury, D., Fraser, M., Masching, R., Nixon, S.A., 2015. What all students in healthcare training programs should learn to increase health equity: perspectives on postcolonialism and the health of aboriginal peoples in Canada. *BMC Med. Educ.* 15 (1), 155. <https://doi.org/10.1186/s12909-015-0442-y>.
- Bell, B., 2021. White dominance in nursing education: a target for anti-racist efforts. *Nurs. Inq.* 28 (1). <https://doi.org/10.1111/nin.12379>.
- Bennett, C., Hamilton, E., Rochani, H., 2019. Exploring race in nursing: teaching nursing students about racial inequality using the historical Lens. *OJIN: the Online J. Issues Nurs.* 24 (2). <https://ojin.nursingworld.org/MainMenuCategories/ANAMarketplace/ANAPeriodicals/OJIN/TableofContents/Vol-24-2019/No2-May-2019/Articles-Previous-Topics/Exploring-Race-in-Nursing.html>.
- Brockman, A.J., 2021. “La crème de la crème”: how racial, gendered, and intersectional social comparisons reveal inequities that affect sense of belonging in STEM. *Sociol. Inq.* 91 (4), 751–777. <https://doi.org/10.1111/soin.12401>.
- Bulfone, G., Maria, M.D., Maurici, M., Macale, L., Sili, A., Vellone, E., Alvaro, R., 2021. Academic failure and its predictors in baccalaureate nursing students: a longitudinal study. *J. Clin. Nurs.* 30 (13–14), 1953–1962. <https://doi.org/10.1111/jocn.15748>.
- Caskill, D., 2012. Discrimination and Public Perceptions of Aboriginal People in Canadian Cities. In: 2012 UAKN Research Paper Series, pp. 1–38. The Urban Aboriginal Knowledge Network. <https://uakn.org/wp-content/uploads/2014/08/2012-UAKN-Research-Paper-Series-Discrimination-and-Public-Perceptions-of-Aboriginal-People-in-Canadian-Cities-Dr.-Don-McCaskill-1.pdf>.
- Castillo, W., Babb, N., 2024. Transforming the future of quantitative educational research: a systematic review of enacting quantCrit. *Race Ethn. Educ.* 27 (1), 1–21. <https://doi.org/10.1080/13613324.2023.2248911>.
- Crenshaw, K., 1991. Mapping the margins: intersectionality, identity politics, and violence against women of color. *Stanford Law Rev.* 43 (6), 1241. <https://doi.org/10.2307/1229039>.
- de Leeuw, S., Greenwood, M., Cameron, E., 2010. Deviant constructions: how governments preserve colonial narratives of addictions and poor mental health to intervene into the lives of indigenous children and families in Canada. *Int. J. Ment. Heal. Addict.* 8 (2), 282–295. <https://doi.org/10.1007/s11469-009-9225-1>.
- Donald, D., 2018. March 28. Canadian education and Indigenous peoples, Open Access News. <https://www.openaccessgovernment.org/canadian-education-and-indigenous-peoples/43884/>.
- Edwards-Maddox, S., Reid, A., Quintana, D.M., 2022. Ethical implications of implicit bias in nursing education. *Reimagining the Future of Nursing Education* 17 (4), 441–445. <https://doi.org/10.1016/j.teln.2022.04.003>.
- Exener-Pirot, H., 2018. Aboriginal Nursing in Canada (No. NOC 2016-301). University of Saskatchewan. https://indigenousnurses.ca/sites/default/files/inline-files/Nursing_AborigNursing_sheet_2018_3.pdf.
- Fine, M., Torre, M.E., Oswald, A.G., Ivory, S., 2021. Critical participatory action research: methods and praxis for intersectional knowledge production. *J. Couns. Psychol.* 68 (3), 344–356. <https://doi.org/10.1037/cou0000445>.
- FitzGerald, C., Hurst, S., 2017. Implicit Bias in healthcare professionals: a systematic review. *BMC Med. Ethics* 18 (1). <https://doi.org/10.1186/s12910-017-0179-8>.
- Gillborn, D., Warming, P., Demack, S., 2018. QuantCrit: education, policy, ‘big data’ and principles for a critical race theory of statistics. *Race Ethn. Educ.* 21 (2), 158–179. <https://doi.org/10.1080/13613324.2017.1377417>.
- González, C.M., Garba, R., Liguori, A., Marantz, P.R., McKee, M.D., Lypson, M.L., 2018. How to make or break implicit bias instruction: implications for curriculum development. *Acad. Med.* 93 (11S), S74–S81. <https://doi.org/10.1097/acm.0000000000002386>.
- Gordon, P., Patterson, L., 2023. Sustaining success for black nursing students. *J. Prof. Nurs.* 49, 102–107. <https://doi.org/10.1016/j.profnurs.2023.09.009>.
- Hawn Nelson, A.L., Zanti, S., 2020. Framework for centering racial equity throughout the administrative data life cycle. *International journal of population data. Science* 5 (3). <https://doi.org/10.23889/ijpds.v5i3.1367>.
- Henry, K.-A.K., Catagnus, R.M., Griffith, A.K., Garcia, Y.A., 2021. Ending the School-to-prison pipeline: perception and experience with zero-tolerance policies and interventions to address racial inequality. *Behav. Anal. Pract.* <https://doi.org/10.1007/s40617-021-00634-z>.
- Henschke, K., 2017. The experiences of indigenous nursing students: a phenomenological study [University of Southern Queensland] <https://pdfs.semanticscholar.org/383c/6dd47676fc93e102fcad9aefc72ca84d4b0a.pdf>.
- Iheduru-Anderson, K., Akanegbu, C., Inungu, J., 2020. Outsiders in nursing - voices of black African born Nurses & Students in the US: an integrative review. *The Open Nursing Journal* 14 (1), 120–134. <https://doi.org/10.2174/1874434602014010120>.
- Iheduru-Anderson, K.C., Agomoh, C.J., Inungu, J., 2021. African born black nurses’ perception of their U.S. work environment: race matters. *Nurs. Outlook* 69 (3), 409–424. <https://doi.org/10.1016/j.outlook.2020.11.009>.
- Lane, A., Petrovic, K., 2018. Educating aboriginal nursing students: responding to the truth and reconciliation report. *Int. J. Nurs. Educ. Scholarsh.* 15 (1). <https://doi.org/10.1515/ijnes-2017-0064>.
- Leon, C.T.T. de, 2020. Attrition to completion progression theory in an associate degree nursing program. *J. Nurs. Educ. Pract.* 10 (5), 11. <https://doi.org/10.5430/jnep.v10n5p11>.
- Lewis, L.S., Willingham, T.L., Milner, A.M.T., 2022. A National Study of progression policies and course repetition in Prelicensure registered nursing programs. *Nurs. Educ. Perspect.* 43 (1), 19–23. <https://doi.org/10.1097/01.NEP.0000000000000862>.
- Luhanga, F., Maposa, S., Puplampu, V., Abudu, E., 2023. “Let’s call a spade a spade. My barrier is being a black student”: challenges for black undergraduate nursing students in a Western Canadian Province. *Can. J. Nurs. Res.* 55 (4), 457–471. <https://doi.org/10.1177/08445621231198632>.
- Manhire-Heath, R., Cormack, D., Wyeth, E., 2019. “...but I just prefer to treat everyone the same...”: general practice receptionists talking about health inequities. *Aust. J. Prim. Health* 25 (5), 430. <https://doi.org/10.1071/PY19026>.
- Oducado, R.M., 2021. Academic performance and the role of self-directed learning, self-esteem, and grit among nursing students. *Jendala Nursing Journal* 5 (1), 1–9. <https://doi.org/10.31983/jnj.v5i1.6634>.
- Ontario Human Rights Commission, 2017, October 23. Under suspicion: concerns about racial profiling in education. Ontario Human Rights Commission <https://www.ohrc.on.ca/en/under-suspicion-concerns-about-racial-profiling-education>.
- Owens, J., McLanahan, S.S., 2020. Unpacking The Drivers Of Racial Disparities In School Suspension And Expulsion. *Social Forces: A Scientific Medium of Social Study and Interpretation* 98 (4), 1548–1577. <https://doi.org/10.1093/sf/soz095>.
- Pacheco-Castillo, J., Casuso-Holgado, M.-J., Labajos-Manzanares, M.-T., Moreno-Morales, N., 2021. Academic stress among nursing students in a private University at Puerto Rico, and its association with their academic performance. *Open Journal of Nursing* 11 (09), 742–756. <https://doi.org/10.4236/ojn.2021.119063>.
- Pijl-Zieber, E.M., Hagen, B., 2011. Towards culturally relevant nursing education for aboriginal students. *Nurse Educ. Today* 31 (6), 595–600. <https://doi.org/10.1016/j.nedt.2010.10.014>.
- Ramsay-Seaner, K., Letcher, A., Isaacson, M., Fenster, K., Heckmann, B., 2022. Assessing color-blind racism in nursing students. *Nurs. Educ. Perspect.* 44 (3), 172–174. <https://doi.org/10.1097/01.nep.0000000000001045>.
- Ray King, K., Fuselier, L., Sirvisetty, H., 2021. LGBTQIA+ invisibility in nursing anatomy/physiology textbooks. *J. Prof. Nurs.* 37 (5), 816–827. <https://doi.org/10.1016/j.profnurs.2021.06.004>.
- Russell, M., Oddleifson, C., Kish, M.R., Kaplan, L., 2022. Countering Deficit Narratives in Quantitative Educational Research. <https://doi.org/10.7275/K44E-SP84>.
- Sedgwick, M., Crosschild, C., Rohatinsky, N., Scott, D.R., 2025. Resilience among indigenous graduate nursing students: a scoping review. *Quality Advancement in Nursing Education - Avancées En Formation Infirmière* 11 (1). <https://doi.org/10.17483/2368-6669.1457>.
- Settles, I.H., Buchanan, N.T., Dotson, K., 2019. Scrutinized but not recognized: (in) visibility and hypervisibility experiences of faculty of color. *J. Vocat. Behav.* 113, 62–74. <https://doi.org/10.1016/j.jvb.2018.06.003>.
- Slattery, S., Cramer, J., Pugh, J.D., Twigg, D.E., 2016. Barriers and enablers to retention of aboriginal diploma of nursing students in Western Australia: an exploratory descriptive study. *Nurse Educ. Today* 42, 17–22. <https://doi.org/10.1016/j.nedt.2016.03.026>.
- Sokolowich, J.R., 2025. Unintended consequences – risk for bias in competency-based assessment. *Nurse Educ. Today* 144, 106404. <https://doi.org/10.1016/j.nedt.2024.106404>.
- Truth and Reconciliation Commission of Canada, author, 2015. Truth and Reconciliation Commission of Canada: Calls to Action. Winnipeg, Manitoba.
- United Nations, 2018. A human rights-based approach to data: leaving no one behind in the 2030 agenda for sustainable development. In: United Nations human rights office of the high commission, pp. 1–24. <https://www.ohchr.org/documents/issues/hri/indicators/guidancenoteonapproachtoadata.pdf>.
- Van Bower, V., 2022. Trauma and survival: the impacts of the COVID-19 pandemic on Indigenous nursing students. *Nurs. Inq.* <https://doi.org/10.1111/nin.12514>.
- Van Bower, V., Woodgate, R.L., Martin, D., Deer, F., 2021. Illuminating Indigenous health care provider stories through forum theater. *AlterNative: An International Journal of Indigenous Peoples* 17 (1), 61–70. <https://doi.org/10.1177/1177180121995801>.
- Vargas, J.H., Peet, J.Z., 2024. The first primer for the QuantCrit-curious critical race theorist or psychologist: on intersectionality theory, interaction effects, and AN(C) OVA/regression models. *J. Soc. Issues* 80 (1), 168–217. <https://doi.org/10.1111/josi.12604>.
- Williams, D.R., Yan, Yu, Jackson, J.S., Anderson, N.B., 1997. Racial differences in physical and mental health: socio-economic status, stress and discrimination. *J. Health Psychol.* 2 (3), 335–351. <https://doi.org/10.1177/135910539700200305>.
- Xu, T., Zhu, P., Ji, Q., Wang, W., Qian, M., Shi, G., 2023. Psychological distress and academic self-efficacy of nursing undergraduates under the normalization of COVID-19: multiple mediating roles of social support and mindfulness. *BMC Med. Educ.* 23 (1). <https://doi.org/10.1186/s12909-023-04288-z>.
- Zanchetta, M.S., Cognet, M., Rahman, R., Byam, A., Carlier, P., Foubert, C., Lagersie, Z., Espindola, R.F., 2021. Blindness, deafness, silence and invisibility that shields racism in nursing education-practice in multicultural hubs of immigration. *J. Prof. Nurs.* 37 (2), 467–476. <https://doi.org/10.1016/j.profnurs.2020.06.012>.